

Echoes of Information Flow: Tracing Volatility and Heavy-tailed Distribution in Cryptocurrency

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Abstract

We examine the role of information flow in shaping volatility and tail risk of cryptocurrency returns. Using a stable-GJR-GARCH model with trading activity as a subordinator, we find that information flow increases conditional volatility while reducing—but not eliminating—heavy tails. Negative news has larger influence on volatility than positive news. The information impact and asymmetric response are stronger for more liquid cryptos. *Trading time* log-returns remain non-Gaussian even after correcting for information-induced heteroskedasticity within *calendar time*, thus challenging Gaussian assumptions in cryptocurrency risk and asset pricing models.

Keywords: Cryptocurrency, Information Flow, Volatility, Heavy Tails, GJR-GARCH

JEL classification: C46, C58, G14, G32

1. Introduction

The recent approval of Bitcoin ETF and Strategic Crypto Reserves underscore the accelerating integration of cryptocurrencies into the mainstream financial system. Unlike conventional assets, cryptocurrencies are traded globally on 24/7 basis, creating a unique and continuous environment for price discovery. In *calendar time*, the unconditional return distribution of cryptocurrencies displays heavy tails, conditional heteroskedasticity and volatility clustering (Tran and Kukal, 2022; Malek et al., 2023). Since the shape of return distribution is influenced by information affecting the intrinsic value and how that information flows to the market (Damodaran, 1985), a crucial question emerges: how does the continuous, global flow of information shape the volatility and tail risk of cryptocurrency returns? Examining this mechanism is paramount for understanding the unique risk dynamics of this emerging asset class.

The subordination process (Clark, 1973) showed that price changes over “*calendar time (time clock)*” reflect an accumulation of new information during a particular “*trading time (volume clock)*”. Due to the randomness of information arrival, the unconditional return distribution presents fat-tails within “*calendar time*”. It is a subordinated stochastic process with a mixture of distributions, in which information flow is the directing process. After adjusting for the information arrival, the logarithm of returns is suggested to follow normal distribution (hereafter log-normal) by Clark (1973), while stable distribution (hereafter log-stable) is recommended by Mandelbrot and Taylor (1967), Granger and Morgenstern (1970) and Mandelbrot (1997). The behavior of crypto return generating process in “*trading time*” remains unclear.

According to Easley et al. (2012), the market makers are now silicon traders that operate on a volume clock, turning over their holdings based on specific numbers of contracts traded rather than specific time intervals. They use a generalized autoregressive conditional heteroskedasticity (GARCH, Bollerslev, 1986) framework to model volatility in the flow of trades. After adjusting to trading time, the impact of volatility clustering reduces. The trading time return distribution is closer to Gaussian distribution and is less heteroscedastic than that in calendar time. Clark’s (1973) subordination process provides theoretical foundation for the GARCH model (see Supplementary Material A). Lamoureux

and Lastrapes (1990, 1994) employ GARCH model to explain the time-varying relationships between information flow and volatility of stocks. They show that the persistence of volatility can be captured by trading volume¹, which serves as a proxy for information flow and acts as a subordinator, creating fat tails and volatility clustering as the GARCH family models. In other words, autocorrelation in the time-varying rate of information arrival causes volatility dependency. However, previous studies using GARCH and subordinated process with Gaussian assumption cannot capture the extreme impact of information on volatility, which could be depicted by stable distribution that meets the Generalized Central Limit Theorem. Previous GARCH family models therefore should be challenged for examining crypto return and volatility.

This study investigates whether the unconditional fat-tailed return distribution is due to volatility persistence created by the random arrival of information. We use a stable-GJR-GARCH model to examine how information asymmetrically influences the extreme changes of cryptocurrencies' return and volatility. We find information flow increases cryptocurrency return volatility but reduces distribution tail heaviness. Negative news raises volatility more than positive news. And stable distribution is more appropriate for describing the log-returns within trading time.

This study makes several contributions: 1) It explains the role of information as a subordinator shaping the fat-tailed distribution of cryptocurrency. We extend Clark's assumption to allow the driving process of information flow to follow stable distribution, which aligns with empirical evidence. 2) It shows that "tails" of cryptos' volatility-adjusted distribution remain significantly fatter than Gaussian distribution. It challenges Gaussian-based crypto risk and asset-pricing models. The findings provide practical implications for diversification, valuation and catastrophic risk detection in crypto markets.

The rest of this paper is organized as follows. Section 2 presents the data and methodology. Section 3 discusses empirical findings. Section 4 concludes the study.

¹ Lamoureux and Lastrapes (1994) relax the assumption of exogenous volume and find that accounting for serial dependence in the information-arrival process does not eliminate GARCH persistence in variance.

2. Data and Methodology

2.1 Data

Our sample comprises all 248 cryptocurrencies in the market with complete time series from 1 January 2019 to 1 January 2025 covering the COVID-19 market wipeout. The closing prices, trading volumes, and market capitalization at daily intervals are retrieved from CoinMarketCap, consisting of 1,635,930 observations. Given higher trading volumes indicate more analyst coverage and larger information flow (Damodaran, 1985), we sort cryptos into five quintiles based on daily trading volume to compare information's influence at different levels. Five value-weighted portfolios are constructed.

2.2 Methodology

Since cryptos exhibit heavy tails and asymmetric volatility effects (Baur and Dimpfl, 2018), we use a “composite” of stable distribution and GJR-GARCH model to examine how information asymmetrically influences the extreme changes of cryptos' return and volatility. Unlike other heavy-tailed distributions, stable distribution meets the Generalized Central Limit Theorem which is crucial for modeling the sum of financial returns. The impacts of information on long-term volatility are captured by the characteristic exponents of stable distribution and thus decrease endogeneity between volatility and volume.

2.2.1 Stable Distribution

The characteristic function $X \sim S(\alpha, \beta, \gamma, \delta)$ of stable distribution proposed by Mandelbrot (1963) and Fama (1965a) can be described as (Nolan, 2014, So, 1987):

$$E[\exp(iuX)] = \begin{cases} \exp(-\gamma^\alpha |u|^\alpha [1 - i\beta(\tan \frac{\pi\alpha}{2})(\text{sign}(\mu))] + i\delta u) & \alpha \neq 1 \\ \exp(-\gamma |u| [1 + i\beta \frac{2}{\pi}(\text{sign}(u)) \times \log(|u|)] + i\delta u) & \alpha = 1 \end{cases} \quad (1)$$

where $i = \sqrt{-1}$ is the imagine number. $\alpha \in (0, 2]$ denotes the characteristic exponent characterizing heaviness of the tails. When $\alpha = 2$, it corresponds to Gaussian distribution. The smaller the α , the heavier the tails. $\beta \in [-1, 1]$ describes skewness. α and β are the shape parameters depicting *long-term tail risk*. $\gamma > 0$ is the scale parameter measuring

dispersion (*short-term volatility*)². $\delta \in \mathbb{R}$ is the location parameter.

2.2.2 Stable-GJR-GARCH

The return in *calendar time* is expressed as (Glosten et al., 1993):

$$r_t = \mu + \varepsilon_t \quad (2)$$

Where μ is the mean return³, $\varepsilon_t \sim S(\alpha, \beta, \gamma, \delta)$ is the stable distributed innovations. The conditional variance h_t is described as (Paoletta, 2016):

$$h_t^\eta = \varpi + \sum_{i=1}^q \lambda_i |\varepsilon_{i,t-i}|^\eta + \sum_{i=1}^o \varphi_i I_{t-i} |\varepsilon_{i,t-i}|^\eta + \sum_{j=1}^p \theta_j h_{j,t-j}^\eta \quad (3)$$

Where $\varpi > 0$ represents the baseline volatility, $I_{t-i} = 1$ if $\varepsilon_{i,t-i} < 0$ (negative news) and 0 otherwise. Parameter λ_i capture the ARCH effect (short-term volatility spikes), θ_j measures the GARCH effect (intermediate volatility persistence), φ_i captures the leverage effect. Coefficients λ_i ($i=0, \dots, p$), φ_i ($i=0, \dots, o$) and θ_j ($j=1, \dots, q$) are all non-negative. The power parameter, η , satisfies $0 < \eta < \alpha$.

When trading volume is added to explain the volatility as in Lamoureux and Lastrapes (1990, 1994), equation (3) takes the form as:

$$h_t^\eta = \varpi + \sum_{i=1}^q \lambda_i |\varepsilon_{i,t-i}|^\eta + \sum_{i=1}^o \varphi_i I_{t-i} |\varepsilon_{i,t-i}|^\eta + \sum_{j=1}^p \theta_j h_{j,t-j}^\eta + \xi v_{t-1} \quad (4)$$

Where v_{t-1} denotes the log-change of volume, ξ represents volume's impact.

3. Empirical Results

3.1 Descriptive Statistics and Heavy Tails

Table 1 shows the descriptive statistics and stable distribution estimates of crypto portfolios. It documents the heavy tails of unconditional return distribution in calendar days (kurtosis > 3 and $\alpha < 2$). As volumes become lower from Portfolio 1 to 5, the size (market capitalization) of portfolios decreases and tails tend to be heavier. It indicates smaller

² Luo (2025) finds that volatility primarily drives the tail exponent. Trading volume significantly affects the scale parameter of extremely large returns in stocks, while its relations with the tail index remains unclear.

³ In GJR-GARCH, μ denotes the expected return, which should not be the time-varying μ_t .

cryptos have higher probability of realizing extreme returns than the larger cryptos.

Table 1. Descriptive Statistics of Crypto Portfolios

	1	2	3	4	5
	HIGH				LOW
Min (r_t)	-0.478	-0.554	-0.512	-0.496	-0.547
Mean (r_t)	0.002	0.002	0.003	0.006	0.012
Median (r_t)	0.002	0.004	0.005	0.004	0.006
Max (r_t)	0.169	0.284	0.245	0.564	1.558
Std (r_t)	0.036	0.042	0.039	0.052	0.083
Skew (r_t)	-1.391	-1.857	-1.558	0.831	7.424
Kurt (r_t)	21.146	23.895	23.040	22.300	121.282
$\alpha(r_t)$	1.543	1.631	1.609	1.460	1.383
$\beta(r_t)$	-0.047	-0.322	-0.172	0.040	0.179
$\gamma(r_t)$	0.017	0.021	0.019	0.021	0.022
$\delta(r_t)$	0.002	0.006	0.006	0.004	0.005
E (VOL_t)	22,626,232,738	15,055,346	2,871,718	341,958	66,682
E (CAP_t)	431,027,491,537	503,872,278	260,532,461	64,081,442	22,755,897
SW (r_t)	0.898	0.878	0.881	0.821	0.542

The portfolios are ranked in descending order based on daily trading volumes. r_t denotes the daily log-returns. E(VOL_t) and E(CAP_t) denote the average daily volume and market capitalization. SW (r_t) denotes the Shapiro-Wilk test statistics

3.2 Information Impact on Volatility and Tail Heaviness

Table 2 compares the parameter estimates of GJR-GARCH model before and after considering information flow under different distribution assumptions. Under Gaussian assumption, Table 2(A2) shows that information flow, proxied by the log-change of volume, significantly increases volatility among all cryptos. The arrival of information (ξ) displays larger impacts on larger size cryptos. All portfolios exhibit significant ARCH and GARCH effects. The leverage effects become significant after accounting for information flow. In Table 2(A1), the sum of coefficients ($\lambda + \theta + 1/2\gamma=0.999$) in Portfolios 4 & 5 is close to 1 as in Integrated-GARCH (Engle and Bollerslev, 1986), which has infinite variance and is non-stationary. In Table 2(A2), the sum of coefficients becomes larger than 1 in Portfolios 1, 4 & 5. The findings show significant impact of information flow, yet Gaussian distribution is inappropriate for modelling the process.

Under stable distribution assumption, Table 2(B) shows significant ARCH, GARCH and leverage effects. Before considering information arrival (Table 2(B1)), all portfolios exhibit pronounced persistence of volatility ($\theta \approx 0.9$). The ARCH effects are significant while lower ($\lambda < 0.1$) than the GARCH effects in all portfolios. Portfolio 1 has the most prominent GARCH effects ($\theta = 0.937$). Compared to the Gaussian assumption (Panel A), the leverage effects (γ) are significant and negative from Portfolio 1 to 4, indicating good

news have larger impact on the conditional volatility in calendar time. While in the smallest Portfolio 5, bad news causes greater volatility as the liquidity is low. The constant terms $\bar{\omega}$ are all significant, indicating inefficient market.

Table 2. Parameter Estimates of Normal- and Stable-GJR-GARCH

	1	2	3	4	5	1	2	3	4	5
Normal-GJR-GARCH										
Panel A1. Without information arrival					Panel A2. With information arrival					
$\bar{\omega}$	0.0001*	0.0001*	0.0000	0.0001	0.0002	0.00001***	0.00010***	0.00005***	0.00004***	0.00003***
	(1.86)	(1.68)	(0.75)	(1.60)	(1.06)	(2.65)	(21627.86)	(120.91)	(3.00)	(256.07)
λ	0.058**	0.118***	0.082***	0.105***	0.231**	0.070***	0.063***	0.044***	0.076***	0.278***
	(2.06)	(3.43)	(3.41)	(3.00)	(2.45)	(188.47)	(1616.96)	(116.07)	(3.49)	(72.23)
θ	0.795***	0.762***	0.875***	0.833***	0.819***	0.863***	0.829***	0.889***	0.878***	0.841***
	(11.45)	(10.45)	(11.48)	(14.85)	(18.54)	(263.17)	(83.69)	(505.14)	(623.88)	(122.52)
φ	0.140	0.113	0.047	0.121	-0.101	0.138***	0.096***	0.057***	0.105***	-0.031***
	(1.31)	(1.06)	(0.50)	(1.27)	(-0.86)	(26.01)	(299.35)	(103.99)	(3.51)	(-59.24)
ξ						0.0011***	0.0009***	0.0009***	0.0008***	0.0004***
						(74.24)	(4898.47)	(120.05)	(8.58)	(40.33)
LLK	4339.39	4052.17	4205.53	3789.40	3024.76	4618.19	4139.72	4334.99	3878.39	3073.35
BIC	-8648.01	-8073.58	-8380.28	-7548.03	-6018.75	-9197.91	-8240.98	-8631.52	-7718.31	-6108.23
Stable-GJR-GARCH										
Panel B1. Without information arrival					Panel B2. With information arrival					
$\bar{\omega}$	0.0002***	0.0009***	0.0005***	0.0004***	0.0006***	0.0091***	0.0005***	0.0004***	0.0004***	0.0006***
	(74.44)	(41.16)	(63.38)	(86.76)	(111.94)	(1309.55)	(36.67)	(123.29)	(76.56)	(129.81)
λ	0.046***	0.078***	0.062***	0.052***	0.032***	0.003***	0.042***	0.032***	0.049***	0.033***
	(272.84)	(148.95)	(171.55)	(188.21)	(190.22)	(10.24)	(80.50)	(174.60)	(171.20)	(211.90)
θ	0.937***	0.867***	0.901***	0.918***	0.927***	0.425***	0.908***	0.930***	0.911***	0.923***
	(2660.46)	(449.76)	(921.36)	(1767.95)	(2187.72)	(929.62)	(613.67)	(2292.67)	(1605.15)	(2338.83)
φ	-0.012***	-0.005***	-0.010***	-0.003***	0.003***	0.169***	0.029***	0.019***	0.015***	0.007***
	(-63.14)	(-7.98)	(-33.38)	(-8.26)	(21.72)	(474.37)	(74.10)	(103.61)	(69.99)	(38.95)
ξ						0.027***	0.015***	0.012***	0.009***	0.005***
						(2020.19)	(549.03)	(653.40)	(329.11)	(363.57)
β	-0.012***	-0.359***	-0.214***	-0.087***	-0.051***	-0.167***	-0.747***	-0.564***	-0.188***	-0.089***
	(-26.22)	(-166.11)	(-118.44)	(-63.88)	(-37.71)	(-78.21)	(-284.21)	(-223.18)	(-132.72)	(-63.27)
α	1.612***	1.752***	1.718***	1.606***	1.532***	1.761***	1.823***	1.809***	1.640***	1.565***
	(502.10)	(334.76)	(381.45)	(563.50)	(685.16)	(318.10)	(301.91)	(283.05)	(493.46)	(637.49)
LLK	4559.63	4273.91	4088.46	3739.28	3745.71	4712.47	4366.63	4522.89	4121.60	3767.70
BIC	-9099.21	-8527.77	-8156.87	-7458.52	-7471.38	-9401.55	-8709.87	-9022.39	-8219.82	-7512.02

The t-statistics in parentheses are measured with robust standard errors. ***, **, * denotes significance at the 1%, 5% and 10% level, respectively. The order (1,1,1) provides better fit for the sample (results available on request).

In Table 2(B2), the inclusion of information flow is significant across all portfolios, which is positively associated with the volatility of unexpected price movement (h_t). The impact of volumes on extreme returns varies with the size of cryptos, which aligns with Westerfield (1977) in stock markets. Figure 1(A) shows similar patterns of conditional volatility and volume. Additionally, after the inclusion of information proxy, innovations

have lighter tails with higher α . The results show that volumes significantly increase conditional volatility but reduce tail heaviness of crypto returns.

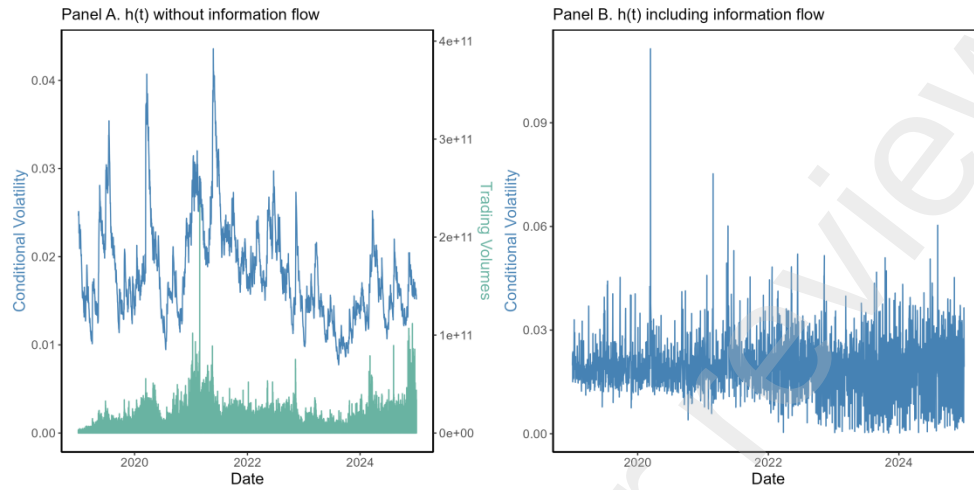


Fig. 1. Conditional Volatility Before and After Including Information Flow (Portfolio 1)

After considering information arrival, Portfolio 1 shows the largest decrease of GARCH effect ($\theta = 0.425$) in Table 2(B) & Fig. 1(B) compared to $\theta = 0.937$ in Table 2(A) & Fig. 1(A), suggesting volume acts as a significant subordinator directing the return generating process of liquid cryptos. For Portfolio 2 and 3, the volatility persistence magnifies. With lower volumes compared to Portfolio 1, medium size cryptos have lower liquidity and news is not quickly reflected by prices, so the information cumulates and volatility spikes decay relatively slowly. For Portfolio 5, the ARCH effect increases after considering information flow, showing stronger influence of short-term volatility spikes when the market is illiquid. According to LeBaron (1992), nontrading and illiquidity could cause serial correlation. With low trading volumes, the behavior of “small cryptos” is like “small firms” suffering from infrequent trading and thus bias systematic risk estimates (Banz, 1981). After accounting for the information impacts, the leverage effects become positive across all portfolios, indicating negative news has larger influence on volatility than positive news.

3.3 Non-Gaussian Distribution in Trading Time

After standardizing information-induced heteroskedasticity, crypto returns and innovations show reduction in tail heaviness (Table 3 & Fig. 2). Larger and more liquid

cryptos demonstrate lighter tails in *trading time*. Portfolio 1 has the lightest tails of standardized innovations with α increasing from 1.543 to 1.872 (Welch's t-test=637.18). However, the characteristic exponents are still significantly less than 2.0. Thus, the log-stable recommended by Mandelbrot and Taylor (1967) is more appropriate than log-normal for describing the returns within both *trading time* and *calendar time*.

Table 3. Stable Distribution Parameter Estimates of Innovations

	1	2	3	4	5
Panel A: Innovations ε_t					
α	1.543*** (-653.15)	1.631*** (537.84)	1.609*** (547.92)	1.460*** (781.99)	1.383*** (903.74)
β	-0.047	-0.322	-0.172	0.040	0.179
γ	0.017	0.021	0.019	0.021	0.022
δ	0.001	0.004	0.002	-0.002	-0.008
Panel B: Standardized Innovations ε_t/h_t					
α	1.872*** (172.61)	1.723*** (331.87)	1.739*** (301.66)	1.621*** (398.86)	1.674*** (341.66)
β	-0.544	-0.843	-0.647	-0.196	0.115
γ	1.131	0.979	0.985	0.972	1.048
δ	0.085	0.299	0.204	0.017	-0.320

The t-statistics are presented in parentheses. The null hypothesis for α to test whether it is significantly different from normal distribution: $\alpha < 2$; ***, **, * denotes significance at the 1%, 5% and 10% level, respectively.

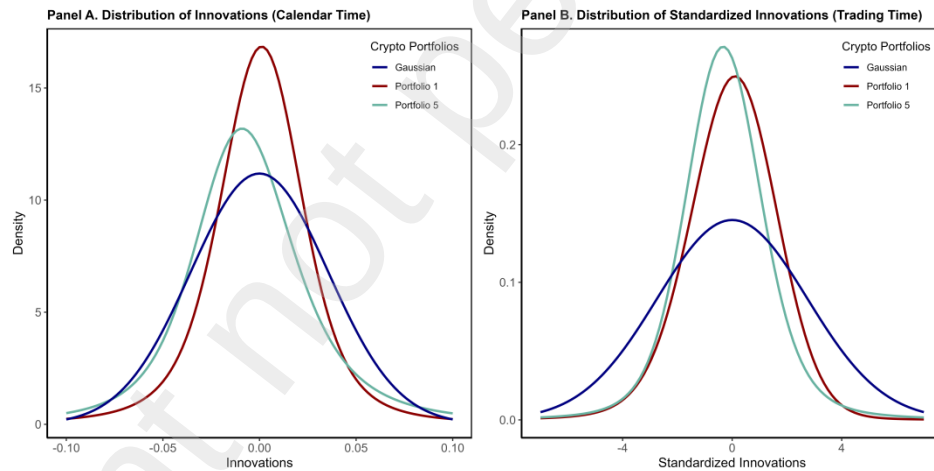


Fig. 2. Distributions in Calendar Time and Trading Time

3.4 Impacts under Information Shocks during COVID-19

Since cryptos are exposed to information shocks when market crash, the impacts during COVID-19 are illustrated in Table 4. Consistent with our main findings, the impacts of volume are significant and positive in all portfolios. Also, the tails become lighter after accounting for information flow. The reduction of tail heaviness and the volume effects both increase with the size of cryptos. During COVID-19, the volume impacts tend to

amplify among all portfolios compared to that in Table 2, showing the market is more sensitive to information during market turbulence. With the inclusion of information flow, the GARCH effects drop dramatically not only in the largest Portfolio 1 but in Portfolio 2, suggesting information shocks during market turmoil enhance its directing role as subordinator. Moreover, the leverage effect (φ) remains positive and increases as the size of cryptos grows. Portfolio 1 displays the largest leverage effect, and φ of Portfolio 2 amplifies from 0.029 to 0.102. It shows during market turbulence, bad news exerts larger influences on the volatility of liquid cryptos compared to their illiquid counterparts.

Table 4. Parameter Estimates of Stable-GJR-GARCH During COVID

	1	2	3	4	5	1	2	3	4	5
Panel A. Without information arrival					Panel B. With information arrival					
ω	0.0005*** (91.25)	0.0015*** (42.30)	0.0014*** (68.54)	0.0006*** (55.62)	0.0004*** (80.66)	0.0081*** (482.64)	0.0037*** (212.04)	0.0012*** (94.19)	0.0009*** (73.76)	0.0005*** (85.5)
λ	0.049*** (180.22)	0.084*** (157.88)	0.079*** (213.89)	0.051*** (110.12)	0.036*** (149.05)	0.046*** (80.86)	0.081*** (216.39)	0.051*** (188.32)	0.064*** (147.20)	0.038*** (177.79)
θ	0.912*** (1735.75)	0.835*** (351)	0.830*** (563.54)	0.907*** (816.17)	0.938*** (1684.46)	0.506*** (629.5)	0.696*** (789.35)	0.857*** (954.89)	0.862*** (707.14)	0.929*** (1853.87)
φ	0.000*** (-6.22)	0.005*** (5.37)	0.017*** (26.30)	-0.002*** (-9.57)	-0.004*** (-16.36)	0.128*** (153.52)	0.102*** (200.85)	0.050*** (104.31)	0.015*** (40.46)	0.003*** (10.81)
ξ						0.027*** (1516.57)	0.019*** (525.1)	0.013*** (464.36)	0.009*** (211.44)	0.006*** (244.67)
β	-0.068*** (-21.26)	-0.448*** (-494.76)	-0.216*** (-84.19)	-0.068*** (-27.41)	-0.192*** (-96.36)	-0.358*** (-75.04)	-0.680*** (-220.16)	-0.601*** (-161.82)	-0.164*** (-84.95)	-0.237*** (-114.11)
α	1.674*** (318.77)	1.786*** (852.23)	1.726*** (279.25)	1.607*** (425.36)	1.650*** (358.01)	1.900*** (138.88)	1.800*** (257.16)	1.812*** (205.98)	1.628*** (390.57)	1.678*** (332.05)
LLK	2322.70	2173.61	2273.30	2277.19	2051.90	2443.99	2213.73	2318.24	2292.81	2070.38
BIC	-4625.35	-4327.17	-4526.55	-4534.33	-4083.76	-4864.59	-4404.07	-4613.09	-4562.22	-4117.38

WHO declares COVID-19 as a public health emergency from 30 January 2020 to 5 May 2023. The t-statistics in parentheses are measured with robust standard errors. ***, **, * denotes significance at the 1%, 5% and 10% level, respectively. The order of GJR-GARCH (1,1,1) provides better fit for the sample (results available on request).

3.5 Robustness Test

The findings are robust to alternative information flow proxies (Supplementary Material B) and other heavy-tailed distributions (Supplementary Material C). We substitute volume with other information proxies, such as the number of transactions (Jones et al., 1994) and the active address (Cong et al., 2021). The results confirm the role of information flow in shaping the return and volatility patterns of cryptos, which is invariant to the choice

of information proxy. And trading time log-returns remain non-Gaussian after considering information-induced heteroskedasticity.

4. Conclusion and Recommendation

This study finds that information flow increases cryptocurrency return volatility but reduces distribution tail heaviness. The directing role of information as a subordinator is stronger for more liquid cryptos. The impact of news on volatility is asymmetric, with negative news generating a larger response on volatility than positive news. During market turmoil, the directing role of information enhances, and the response from bad news amplifies among liquid cryptos. In addition, our findings demonstrate that stable distribution is more appropriate for describing the log-returns within both *trading time* and *calendar time*.

It is recommended to 1) For Portfolio Management: Shift the objective from minimizing variance to actively managing tail risk in crypto markets. According to Fama (1965b), the diversification effect is severely weakened under a stable distribution framework. The portfolios could be supplemented with hedging or alternative optimization strategies. 2) For Asset Pricing and Risk Management: The confirmed presence of heavy tails implies a substantially higher probability of extreme losses and persistent fluctuations than assumed under Gaussian distribution. The crypto valuation framework by Cong et al. (2021) could be extended to allow for the impact of heavy tails. 3) For Catastrophic Risk Detection: The information proxy in this study can be developed into timely indicators of market turmoil. Following the approach of Easley et al. (2012) in analyzing the ‘Flash Crash’, our model offers a foundation in constructing early warning systems for future crises in crypto markets.

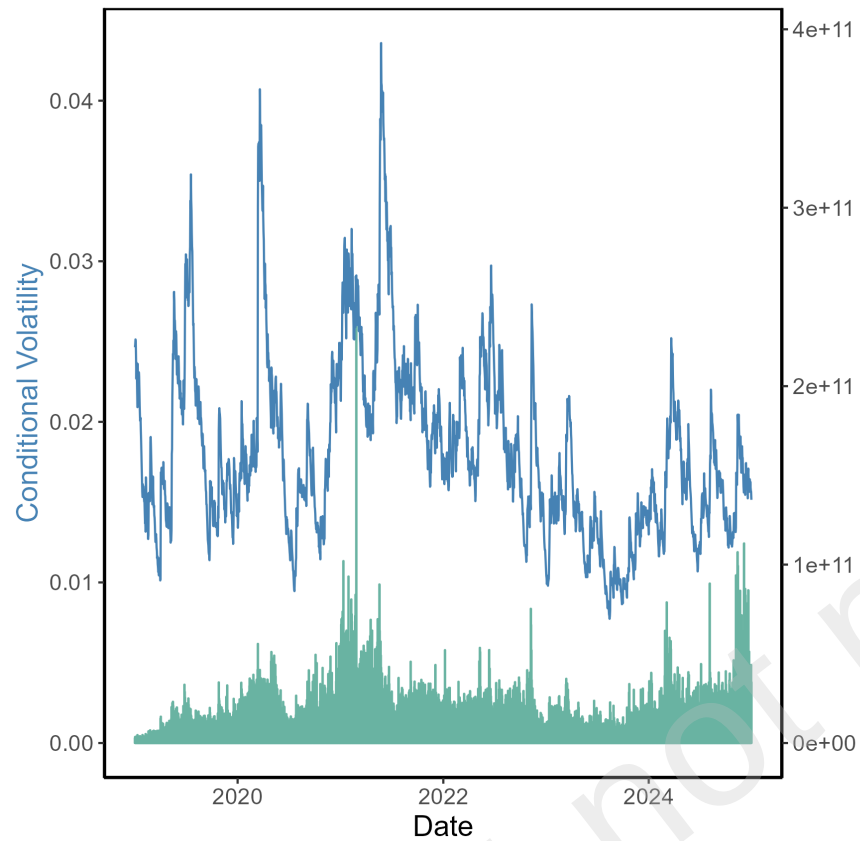
The potential limitation of this study lies in the noises of high-frequency trading and market microstructure, where trading volume can be contaminated by non-informed trading (Feng et al., 2018). Given the limited time span of high-frequency data for a broad cross-section of cryptos, a more detailed investigation will be conducted in future research.

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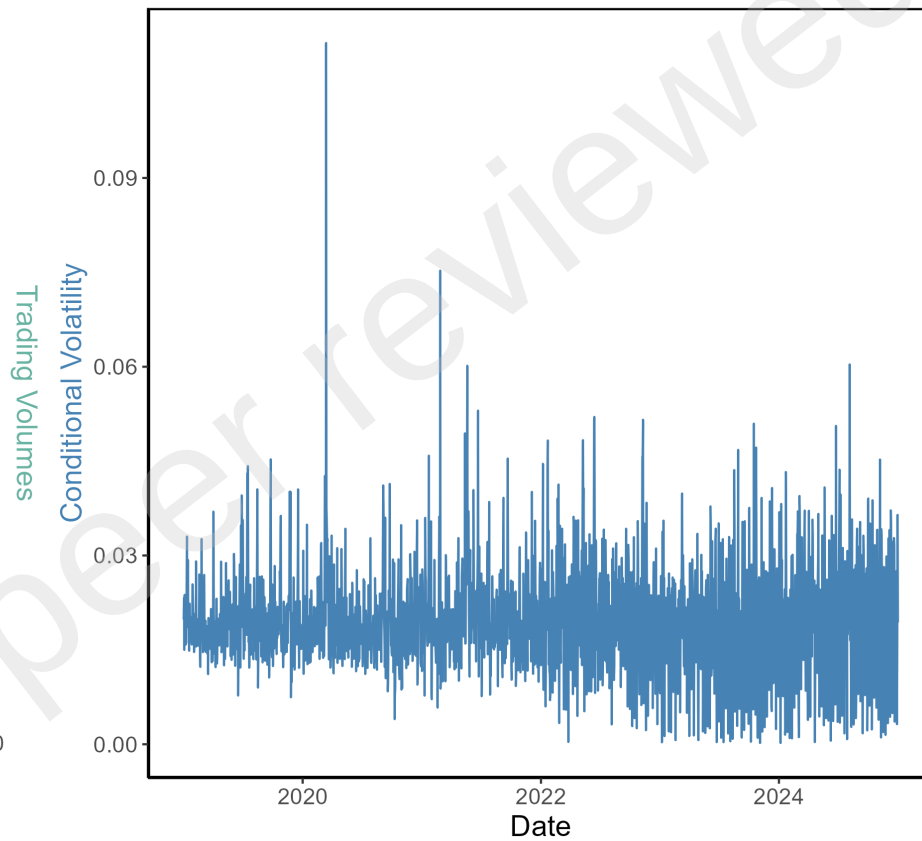
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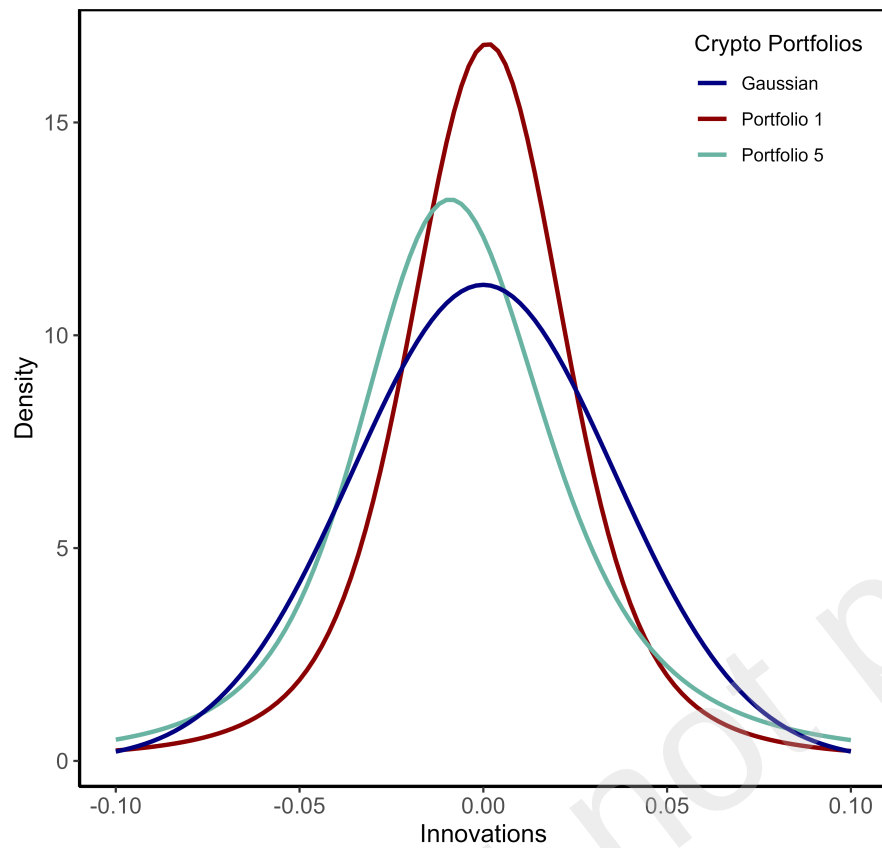
Panel A. $h(t)$ without information flow



Panel B. $h(t)$ including information flow



Panel A. Distribution of Innovations (Calendar Time)



Panel B. Distribution of Standardized Innovations (Trading Time)

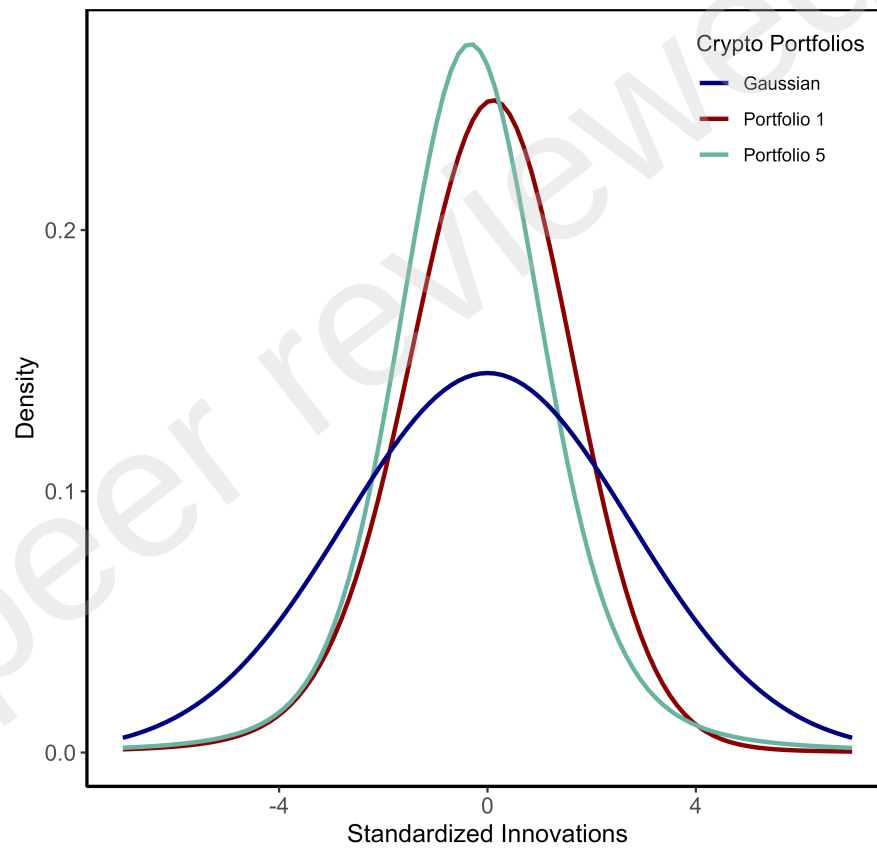


Figure Captions

Fig. 1. Conditional Volatility Before and After Including Information Flow (Portfolio 1)

Fig. 2. Distributions in Calendar Time and Trading Time